

MATHEMATICS
Introduction to Modern Algebra II
Quiz 6.

Give a detailed proof of the following fact: Let F be a field and $f(x) \in F[x]$ be a polynomial of degree n . There is an extension K of F of degree at most $n!$ such that $f(x)$ splits in to linear factors in $K[x]$.

Proof. If $\deg f(x) = 1$, then let $K = F$. Clearly $f(x)$ splits in to linear factors in $F[x]$ and $[F : F] = 1$. Proceeding by induction, we may assume that the claim holds for all polynomials of degree $\leq n - 1$.

We begin by showing that if $\deg f(x) = n$, then there is an extension $F_1 \subset F$ containing a root of $f(x)$ such that $[K_1 : F] \leq n$. To see this let $q(x) \in F[x]$ be an irreducible factor of $f(x)$. Then $K_1 = F[x]/(q(x))$ is a field, and $a_1 = x + (q(x))$ is a root of $q(x)$ in K_1 and hence also a root of $f(x)$ in K_1 (it suffices to show that $q(a_1) = 0$; but $q(a_1) = q(x) + (q(x)) = (q(x)) = 0 \in K_1 = F[x]/(q(x))$). Therefore, we can factor $f(x) = f_1(x)(x - a_1)$ in $K_1[x]$.

By induction on the degree, there is an extension $[K : K_1] \leq (n - 1)!$ such that $f_1(x)$ splits in to linear factors in $K[x]$ and hence so does $f(x)$. Finally note that $[K : F] = [K : K_1][K_1 : F] \leq n \cdot (n - 1)! = n!$. $\square$